

Biological Clues to Potent DNA-Damaging Activities in Food and Flavoring

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Population differences in age-related diseases and cancer could stem from differences in diet. To characterize DNA strand-breaking activities in selected foods/beverages, flavorings, and some of their constituent chemicals, we used p53R cells, a cellular assay sensitive to such breaks. Substances testing positive included reference chemicals: quinacrine (peak response, 51X) and etoposide (33X); flavonoids: EGCG (19X), curcumin (12X), apigenin (9X), and quercetin (7X); beverages: chamomile (11X), green (21X), and black tea (26X) and coffee (3 to 29X); and liquid smoke (4 to 28X). Damage occurred at dietary concentrations: etoposide near 5 $\mu\text{g/ml}$ produced responses similar to a 1:1000 dilution of liquid smoke, a 1:20 dilution of coffee, and a 1:5 dilution of tea. Pyrogallol-related chemicals and tannins are present in dietary sources and individually produced strong activity: pyrogallol (30X), 3-methoxycatechol (25X), gallic acid (21X), and 1,2,4-benzenetriol (21X). From structure-activity relationships, high activities depended on specific orientations of hydroxyls on the benzene ring. Responses accompanied cellular signals characteristic of DNA breaks such as H2AX phosphorylation. Breaks were also directly detected by comet assay. Cellular toxicological effects of foods and flavorings could guide epidemiologic and experimental studies of potential disease risks from DNA strand-breaking chemicals in diets.